

AREC 615 Problem Set: Linear Programming

Show your work and justify your answers. Complete Questions 1, 3, and 4 on paper; software checks are optional. Use R or Excel Solver for Question 2 and append your code or spreadsheet setup.

1 Farm Allocation and Sensitivity (On Paper)

A farm grows **tomatoes** and **bell peppers**, earning \$60 and \$40 per acre, respectively. It has **8 acres** and **10 irrigation units** available. Tomatoes use 2 irrigation units per acre; peppers use 1. Acreages may be fractional.

Let x_1 and x_2 denote tomato and pepper acres. For each experiment, start from this baseline and change only the stated parameter.

- Formulate.** Write the LP in scalar and matrix form, including nonnegativity. Define x , c , A , and b , label the constraints, and state the units of the objective and constraint coefficients.
- Solve.** Graph the feasible region and an isoprofit line. Find the optimal vertex, verify it algebraically, and calculate profit. Identify binding constraints and unused resources.
- Follow the simplex logic.** Add slack variables and start at the origin with tomatoes entering first. Use the ratio test to identify the leaving variable and next vertex. Along the binding irrigation constraint, determine whether introducing peppers raises profit. Find the next vertex and justify optimality. Identify the basis at the origin and after each move. **No tableau is required.**
- Value the resources.** Compute and interpret both baseline shadow prices, including their units. Verify irrigation's value by increasing availability from 10 to 11 units.
- Change a profit coefficient.** Hold pepper profit at \$40. Derive the tomato-profit range over which the baseline plan remains optimal. Explain the endpoint solutions and identify the optimal plan just outside each end. Does an unchanged planting plan imply unchanged profit?
- Check a prediction's limit.** With land fixed at 8 acres, derive the irrigation range over which the baseline basis remains optimal. Predict the profit gain from increasing irrigation from 10 to 18 units using its baseline shadow price, then solve the changed model. Explain any discrepancy and change in binding constraints.
- Evaluate a shortcut.** A colleague claims irrigation's shadow price equals $60/2 = 30$, tomato profit divided by irrigation use per acre. Explain why this fails using the changes in both acreages when irrigation increases slightly. State conditions under which such a ratio correctly values an additional resource unit.

2 Distribution Optimization (Use Computer)

A food distribution network has **2 warehouses** (W1 and W2) and **5 cities** (C1 through C5). Each warehouse can ship to every city. Choose weekly shipments to meet minimum city demands at minimum cost while respecting warehouse capacities. Shipments may be fractional.

Shipping costs are in **dollars per unit**:

Warehouse	C1	C2	C3	C4	C5
W1	4	6	8	7	5
W2	5	4	7	6	6

Weekly minimum demands are **80, 70, 60, 90, and 50 units** for C1 through C5, respectively. Weekly warehouse capacities are **200 units at W1** and **180 units at W2**.

- Formulate.** Let x_{ij} denote units shipped from warehouse i to city j per week. Write the objective, the seven structural constraints (two capacity limits and five demand requirements), and the nonnegativity restrictions. Explain your inequality directions.
- Solve and verify.** Use R (for example, `lpSolve`) or Excel Solver. Report an optimal shipment plan and weekly cost. Independently reconstruct the cost and verify all constraints from the reported shipments. Provide a table with each constraint's direction, right-hand side, left-hand side at the optimum, slack or surplus, and shadow price.
- Interpret and predict.** Explain the city-demand and warehouse-capacity shadow prices to a nontechnical manager. Define a shadow price as the change in minimized weekly cost per unit increase in a constraint's right-hand side. Predict the cost saving from adding 10 units of capacity at W1 and, separately, from adding 10 units at W2. Re-solve each scenario and recommend which expansion is more valuable before paying for added capacity.
- Find the limit of the prediction.** Starting from the baseline again, increase W2 capacity by 50 units. Compare the actual saving with the baseline shadow-price prediction. Determine how much W2 capacity can increase before the marginal saving changes, and explain the economic reason for that change. A full ranging table for every coefficient is not required.
- Check uniqueness.** Determine whether the optimal shipment plan is unique. If it is not, construct a second optimal plan and verify feasibility and equal cost. Explain what the shipping-cost differences imply for reallocating shipments to C3 and C4 between warehouses.

Solver guidance. For a $\leq$ constraint, slack is RHS minus LHS; for a $\geq$ constraint, surplus is LHS minus RHS. Identify binding constraints from equality of LHS and RHS, allowing for numerical tolerance, rather than from the sign of a shadow price. The farm R example's sensitivity-table code assumes a maximization model with $\leq$ resource constraints and needs adaptation here. Check shadow-price signs against the cost changes from your re-solves.

3 Dual Formulation and Interpretation (On Paper)

Consider the following primal LP. Interpret x_1 and x_2 as activity levels, the objective coefficients as profits per activity unit, and the two constraints as resource limits.

$$\begin{array}{ll}\max & 100x_1 + 150x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 10 \\ & 3x_1 + x_2 \leq 12 \\ & x_1, x_2 \geq 0.\end{array}$$

- a. **Formulate the dual.** Define its variables as per-unit resource valuations. Interpret the dual objective and explain why each dual constraint requires the value of an activity's resource bundle to cover its profit.
- b. **Solve both problems.** Find the optimal primal activity levels and dual resource values algebraically. Verify primal and dual feasibility and calculate both objective values. You may use software to check your calculations.
- c. **Explain the duality results.** Verify strong duality numerically. Explain how weak duality makes any feasible dual valuation an upper bound on attainable primal profit, and why equal objective values certify optimality when both solutions are feasible.
- d. **Check complementary slackness.** Calculate the slack in each primal and dual constraint and verify both sets of complementary-slackness conditions. Relate resource use to resource values and activity levels to the values of their resource bundles.
- e. **Create a contrasting case.** Reduce the profit coefficient of x_1 from 100 to 50, keeping all other coefficients unchanged. Solve the modified primal and dual. Identify any unused activity and slack resource. Use complementary slackness to explain the zero resource price and why the unused activity's resource opportunity cost exceeds its profit. Verify that the two optimal objective values still agree.

4 Valuing a Land Package (On Paper)

In the Stewart and Greenhalgh application, a planner minimizes agricultural production cost while meeting fixed output requirements. An additional acre of **group 2 land** changes four right-hand sides together. At the 100-acre baseline, the relevant restrictions are

$$\begin{aligned} C_2 + W_2 + P_2 &\leq 100 && \text{(total land),} \\ C_2 &\leq 40 && \text{(corn maximum),} \\ W_2 &\leq 30 && \text{(wheat maximum),} \\ P_2 &\geq 30 && \text{(pasture minimum),} \end{aligned}$$

where C_2 , W_2 , and P_2 are acres of corn, wheat, and pasture. All other model parameters remain fixed. Use the following valid set of baseline cost shadow prices:

Constraint	Cost shadow price	RHS change per added acre
Total land	−1.14917	1.0
Corn maximum	−1.55083	0.4
Wheat maximum	0	0.3
Pasture minimum	+1.44250	0.3

Each cost shadow price measures the change in minimized production cost per unit increase in that row's RHS. Monetary values refer to the model's accounting period. Individual multipliers can be nonunique; use the supplied values for this calculation.

- Value the coordinated change.** Write the four restrictions after an expansion of t acres. Derive the local change in minimized cost per added acre as a weighted sum of shadow prices, then calculate the marginal production-cost saving. Explain the sign difference between a cost change and a cost saving.
- Interpret the package.** Explain why the total-land shadow price alone does not value this expansion, and why the pasture requirement offsets some benefits. The wheat maximum binds at the baseline but has a zero multiplier in the supplied solution. Does binding necessarily imply a nonzero shadow price? Explain.
- Value a finite project.** A planner proposes multiplying the baseline marginal saving by 20 to value a 20-acre expansion. State when that calculation is valid and explain how to calculate the gross production-cost saving if the marginal value changes during the expansion. Describe which bounds must change together when re-solving the model. Explain how you would compare gross savings with development costs measured over the same accounting period.